

Field Deployment Results: 145,000+ Tonnes Across Indian Smallholder Farms

Executive Summary

We have access to and have forensically analyzed paired crop-cut data from 216 smallholder rice farmers across 42 villages in central and eastern India. The deployment programme applied crushed basalt sourced from two major geological formations (the Deccan Traps and the Rajmahal Traps) across 6,937 hectares during the 2025 Kharif season. Total basalt deployed: 145,157 tonnes.

What We Have

- Per-farmer paired crop-cut observations (n = 216) including baseline (control) yield, treatment yield, percentage yield change, and farm size
- Site-specific deployment parameters including application rates (t/ha) and total tonnage
- Full economic analysis including MSP-adjusted income calculations across three Indian states

Site-Level Results

Deployment Zone	N (Farmers)	Mean Yield Increase	Median Yield Increase	Std Dev	Area (ha)	Application Rate (t/ha)	Basalt Deployed (t)
Zone A (Central India)	85	+24.6%	+23.5%	10.6%	3,068	16.4	50,427
Zone B (Central India)	83	+38.4%	+34.6%	38.2%	2,204	25.7	56,634
Zone C (Eastern India)	32	+13.3%	+12.2%	6.6%	1,131	25.3	28,566
Zone D (Central India)	10	+33.2%	+25.0%	24.1%	317	18.1	5,755
Zone E (Central India)	6	+54.7%	+56.4%	18.3%	217	17.4	3,775
Total	216	+29.5%	+24.7%	27.1%	6,937	—	145,157

The “Poor Soil Premium” Discovery

Our most significant finding: There is a strong inverse correlation ($r = -0.41$) between baseline soil productivity and yield response.

- **Lowest productivity quartile (Q1):** Average yield gain of +47%
- **Highest productivity quartile (Q4):** Average yield gain of +18%
- **Farm size shows no correlation** with yield response ($r = -0.01$), meaning benefits scale proportionally across all holding sizes

This means ERW disproportionately benefits the most resource-constrained farmers on the most degraded soils — the exact demographic most underserved by conventional fertilizer pathways.

Economic Impact (Per-Farmer)

- Median yield gain: 3.29 quintals/acre
- Net income uplift per farmer per season: 207–443 (depending on state procurement prices)
- Sample-weighted average income uplift: ~\$355/season (~22% increase in annual agricultural household income)
- **Farmer acquisition cost: \$0** (basalt provided at no cost; costs recovered through carbon finance)

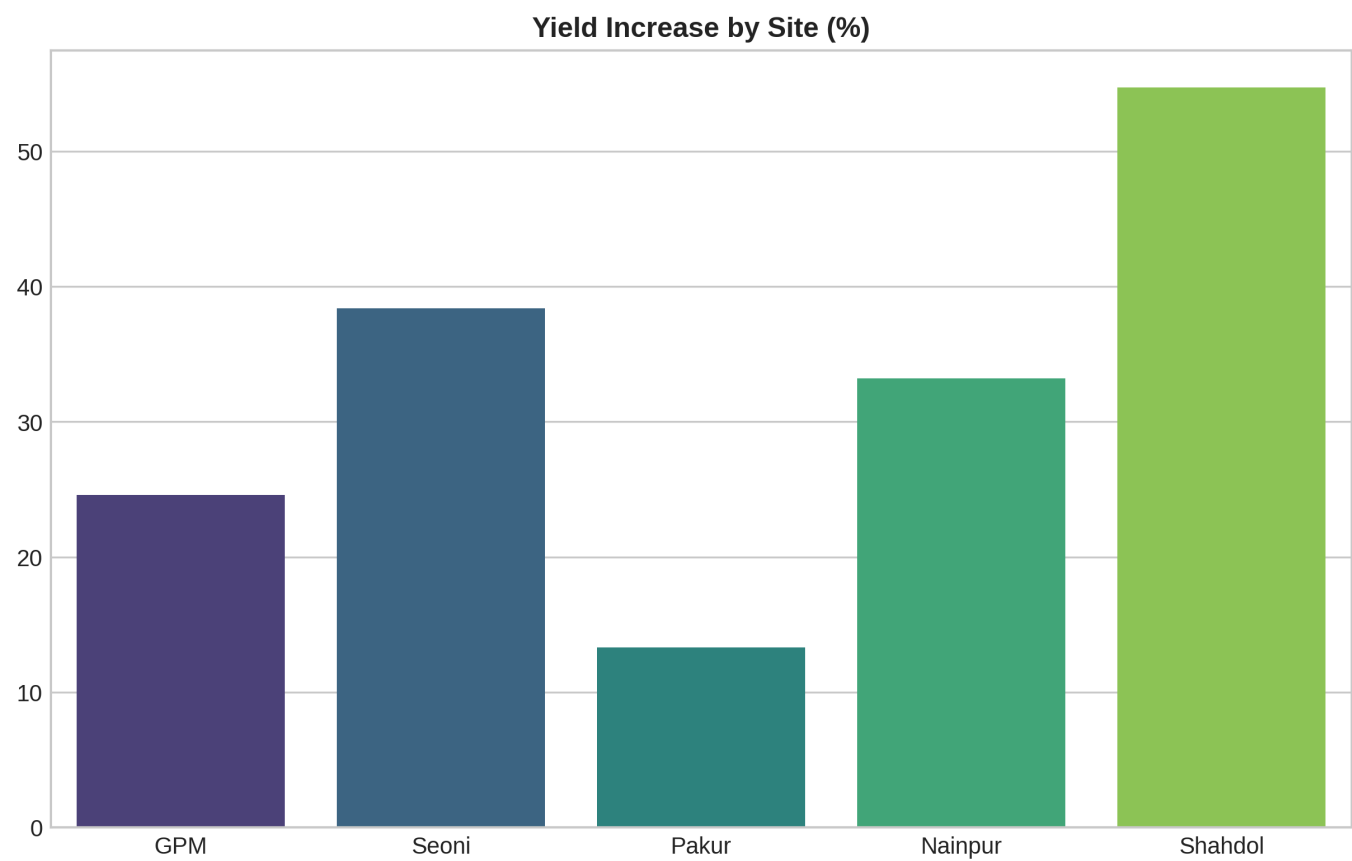
What This Proves

1. ERW delivers statistically significant yield gains across diverse geologies (Deccan Traps vs. Rajmahal Traps)
2. The yield response is inversely correlated with baseline soil quality — targeting degraded soils maximizes both agronomic and carbon outcomes
3. The economics work without requiring the farmer to pay anything — carbon finance covers deployment costs
4. These are not greenhouse results. This is real-world, at-scale data from 216 commercial farming operations

What We Are Not Sharing Here

- Exact village-level GPS coordinates of deployment sites
- The specific identity of the deployment partner
- Per-farmer line-item economic calculations
- The proprietary geographic targeting algorithm we built from this data

Yield Analysis by Site



Yield Analysis by Soil Quality

